

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL A-LEVEL BIOLOGY (9610)

Unit 5 Synoptic paper

Thursday 18 January 2024 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 75.
- In Question 6, 2 marks will be awarded for the quality of your written communication. You will be marked on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
TOTAL	



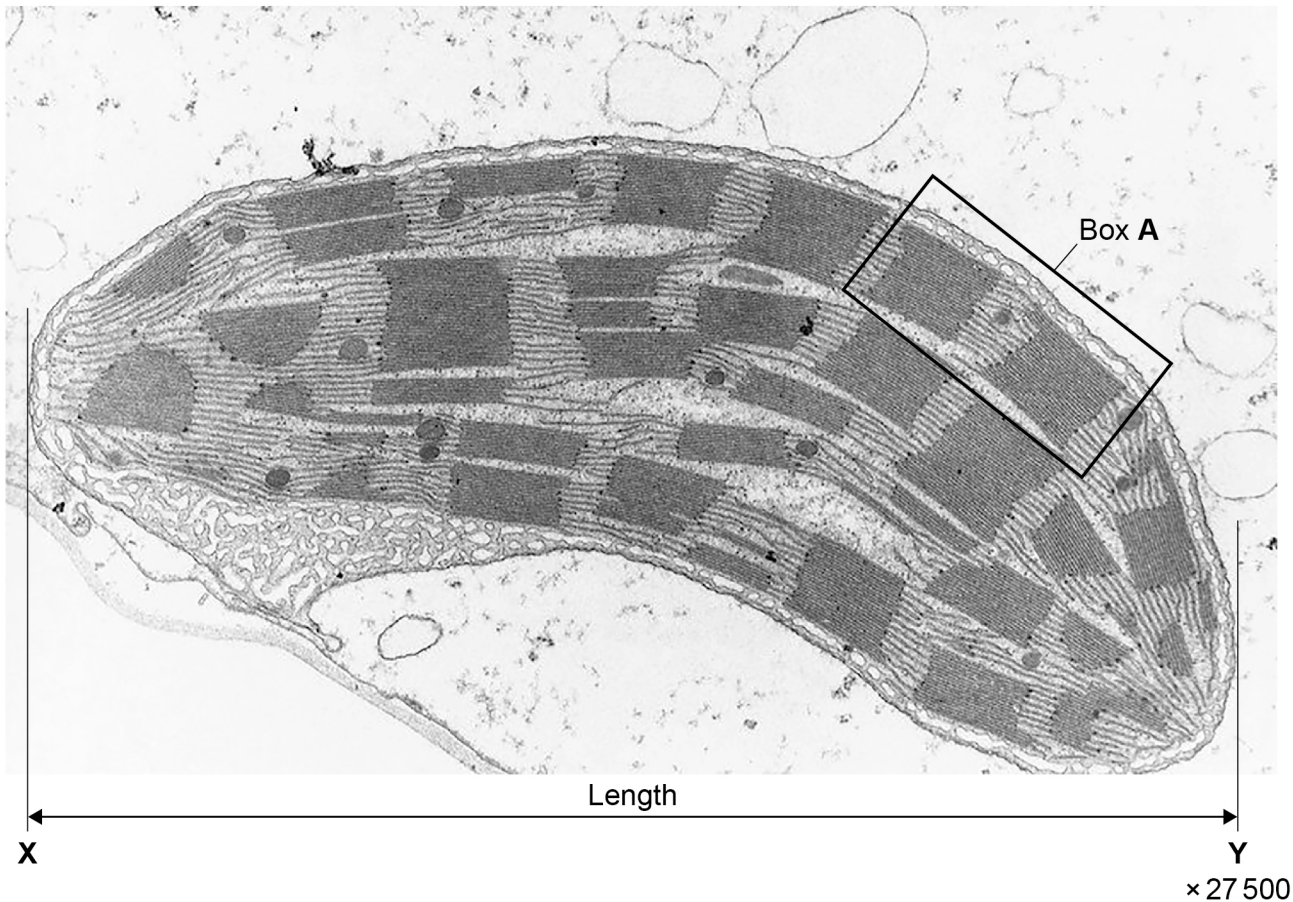
Answer **all** questions in the spaces provided.

0 1

Chloroplasts are cell organelles found in many eukaryotes.

Figure 1 shows a chloroplast from the leaf of a corn plant.

Figure 1



0 1 . 1

Scientists produced the image in **Figure 1** using a transmission electron microscope (TEM).

Give **two** pieces of evidence from **Figure 1** that a TEM was used.

[2 marks]

1 _____

2 _____



0	1	.	2
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Draw a large, clear diagram of the structures in box **A** in **Figure 1**.

[3 marks]

0	1	.	3
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The magnification of **Figure 1** is $\times 27\,500$

Calculate the length, **X–Y**, of the actual chloroplast.

Give your answer in micrometres.

[2 marks]

Length of the actual chloroplast, **X–Y** = _____ μm

Question 1 continues on the next page

Turn over ►



Students investigate the **light-dependent** reaction of photosynthesis.

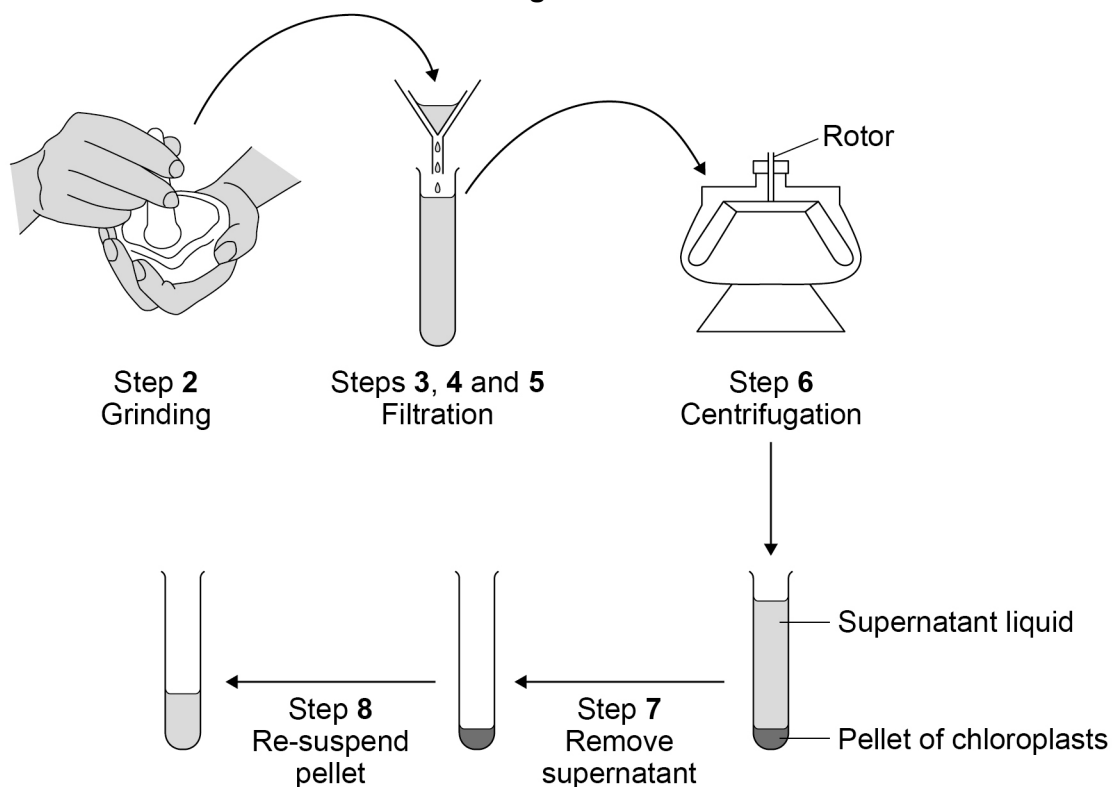
The students first make a preparation of chloroplasts.

This is the method the students use.

1. Cut 3 small leaves into pieces with scissors.
2. Put the pieces of leaf into a mortar at 4 °C, add 20 cm³ of a cold isolation solution and grind the pieces of leaf with a pestle.
3. Put 4 layers of thin cloth in a funnel and wet the cloth with cold isolation solution.
4. Put several centrifuge tubes in a beaker of ice + salt + water and filter the mixture from step 2 into each centrifuge tube.
5. Check that each centrifuge tube contains the same volume of filtrate.
6. Spin the centrifuge tubes for 10 minutes in a bench centrifuge to give a green pellet of chloroplasts in each tube.
7. Pour off the supernatant liquid from each centrifuge tube.
8. Re-suspend the pellet of chloroplasts by mixing with 2 cm³ of cold isolation solution and store in a beaker of ice + salt + water.

Figure 2 gives a summary of the students' method.

Figure 2



The isolation solution (steps **2**, **3** and **8**) is a buffer solution at pH 7.0 and contains 0.4 mol dm^{-3} sucrose.

0 1 . 4

Give the reason why the students use a solution at pH 7.0 for isolating the chloroplasts.

[1 mark]

0 1 . 5

The 0.4 mol dm^{-3} sucrose solution has a water potential only slightly lower than the stroma inside the chloroplasts.

Explain why the students included 0.4 mol dm^{-3} sucrose in the isolation solution.

[2 marks]

0 1 . 6

Explain why the students check that each centrifuge tube contains the **same** volume of filtrate (step **5**) before placing the tubes in the centrifuge.

[2 marks]

Question 1 continues on the next page

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[4 marks]

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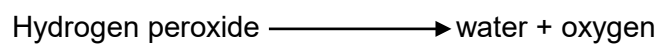
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0 2

Students investigate the effect of substrate concentration on the activity of the enzyme catalase.

Catalase catalyses the reaction:

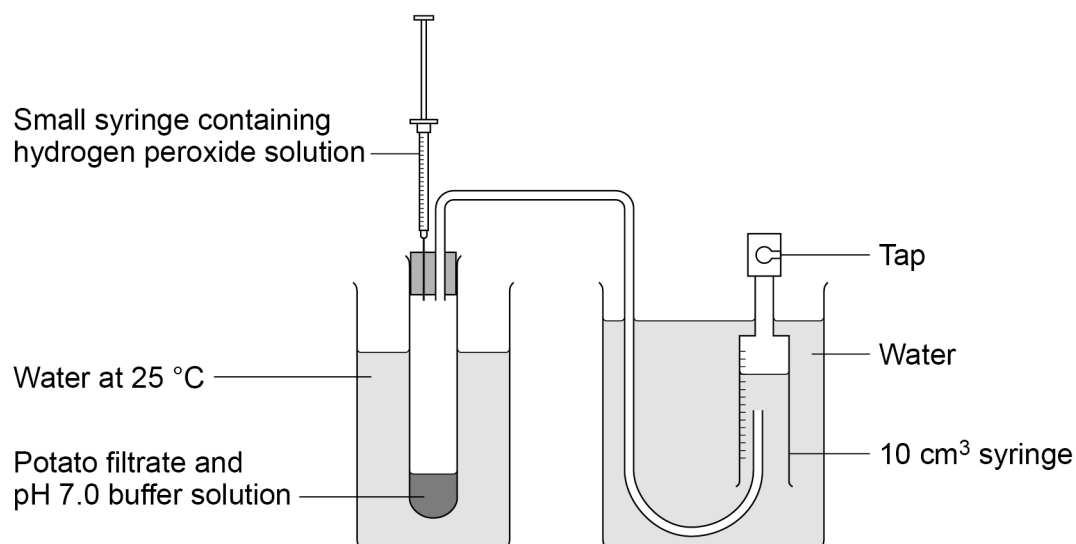


The students:

- grind up 150 grams of potato tissue in 100 cm³ of water
- filter the mixture and use the filtrate as their source of catalase
- collect the oxygen produced in the reaction in a 10 cm³ syringe.

Figure 3 shows the equipment the students use.

Figure 3



Question 2 continues on the next page

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ANSWER IN THE SPACES PROVIDED**

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Table 2 shows the students' results.

Table 2

Concentration of hydrogen peroxide / %	Mean volume of oxygen collected $\pm$ SD / cm ³
100	9.6 $\pm$ 0.24
80	9.5 $\pm$ 0.12
60	9.2 $\pm$ 0.62
40	8.0 $\pm$ 0.61
20	4.6 $\pm$ 0.36
10	2.4 $\pm$ 0.28
0	0.0 $\pm$ 0.00



Describe in detail how the students could use the equipment shown in **Figure 3** to obtain the results shown in **Table 2**.

[illegible]

Turn over ►



0 2 . 2

Describe how the students should display their results in a graph.

Give a reason for your choice of graph.

[3 marks]

0 2 . 3

One student states:

'There is no significant difference between the results with 40% hydrogen peroxide and 60% hydrogen peroxide.'

Use data from **Table 2** to give evidence to support the student's statement.**[2 marks]**



0 3

Scientists can produce fragments of DNA using different methods.

0 3 . 1

In one method, the scientists produce complementary DNA (cDNA) from mRNA.

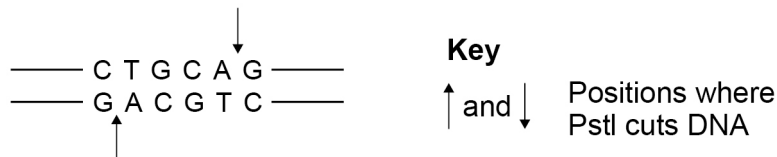
Name the type of enzyme that produces cDNA from mRNA.

[1 mark]

In another method, the scientists use an enzyme that cuts DNA into smaller fragments.

An enzyme called PstI cuts DNA at the specific base sequence shown in **Figure 4**.

Figure 4



0 3 . 2

What general name is given to enzymes such as PstI?

[1 mark]

0 3 . 3

PstI produces DNA fragments with sticky ends.

Why are sticky ends useful in genetic engineering?

[1 mark]

Question 3 continues on the next page

Turn over ►



The scientists prepare a sample containing many identical molecules of DNA.

The DNA molecules are linear (non-circular).

The scientists treat a sample of the DNA with PstI (**Sample 1**).

The scientists leave a second sample of the DNA untreated (**Sample 2**).

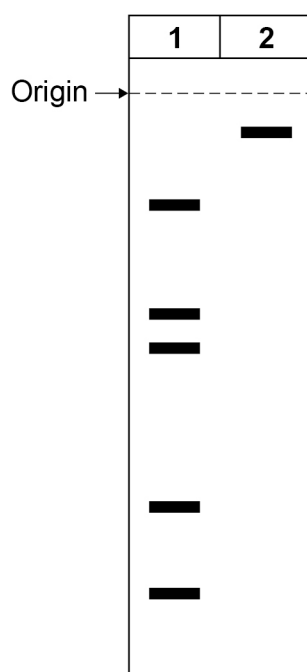
The scientists carry out gel electrophoresis on each of the two samples.

Gel electrophoresis separates pieces of DNA depending on their electrical charge and their mass.

Small pieces of DNA move further from the origin than large pieces of DNA.

Figure 5 shows the results.

Figure 5



Key

Sample 1 = DNA treated with PstI

Sample 2 = Untreated DNA

0 3 . 4

How many times did the base sequence **CTGCAG** occur in the untreated DNA?
GACGTC

[1 mark]



0 3 . 5

In a certain genetic condition, **one** of the **CTGCAG** sequences is changed.
GACGTC

Predict how this would change the appearance of **Sample 1** in **Figure 5**.

[2 marks]

6

Turn over for the next question

Turn over ►



0 4

In domestic cats, some fur colours are controlled by a gene on the X chromosome.

A scientist crosses black cats with orange cats.

The results are different depending on which parent is black and which is orange.

Cross 1: Black male cats are crossed with orange female cats. All the male offspring are orange and all the female offspring are 'tortoiseshell' (a mixture of black and orange fur).

Cross 2: Orange male cats are crossed with black female cats. All the male offspring are black and all the female offspring are, again, tortoiseshell.

In Questions **04.1** and **04.2**, use the following symbols:

X^B = an X chromosome with the allele for black fur

X^b = an X chromosome with the allele for orange fur

Y = a Y chromosome.

0 4 . 1

Complete the **two** genetic diagrams to explain the scientist's results.

[4 marks]**Cross 1****Black male × orange female**Parental
genotypes

Gametes

Offspring
genotypes

Offspring
phenotypes

Offspring
sex

Cross 2**Orange male × black female**Parental
genotypes

Gametes

Offspring
genotypes

Offspring
phenotypes

Offspring
sex



0 4 . 2

The scientist then crosses orange male cats with tortoiseshell female cats.

Table 3 shows the offspring.

Table 3

Black male	Orange male	Orange female	Tortoiseshell female
12	7	13	8

The scientist expected a 1 : 1 : 1 : 1 ratio of phenotypes.

The scientist uses the χ^2 (chi-squared) test to analyse the results.

The scientist calculates a value of $\chi^2 = 2.60$

Table 4 gives probability values for different values of χ^2

Table 4

Degrees of freedom	Probability value					
	0.99	0.95	0.10	0.05	0.01	0.001
1	0.002	0.004	2.71	3.84	6.63	10.83
2	0.020	0.103	4.61	5.99	9.21	13.82
3	0.115	0.352	6.25	7.81	11.34	16.27
4	0.297	0.711	7.78	9.49	13.28	18.47

Explain if the scientist's results in **Table 3** fit the expected ratio.

Use the calculated value of $\chi^2 = 2.60$ and data from **Table 4**.

[3 marks]



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0 5

Leukaemia is a cancer of the blood cells.

Vincristine and resveratrol are two drugs that might be useful in treating leukaemia.

- Vincristine – an inhibitor of microtubule formation.
- Resveratrol – an antioxidant.

0 5 . 1

Microtubules are components of spindle fibres.

Explain how an inhibitor of microtubule formation could be useful in cancer treatment.

[2 marks]

0 5 . 2

An antioxidant is a chemical that prevents free radical formation.

Free radicals can damage DNA.

Suggest **one** way an antioxidant might help to prevent cancer.

[1 mark]

Question 5 continues on the next page

Turn over ►

Scientists investigate the effects of vincristine and resveratrol on one type of leukaemia.

The scientists put human leukaemia cancer cells into 48 mice.

The scientists then divide the mice into 3 groups of 16 mice.

The scientists treat the mice once every 2 days for 4 weeks, as follows:

Group 1: inject the mice with vincristine (0.5 mg kg^{-1} body mass)

Group 2: inject the mice with resveratrol (10.0 mg kg^{-1} body mass), dissolved in dimethyl sulfoxide (DMSO)

Group 3: inject the mice with DMSO only.

0 5 . 3

The scientists inject a certain mass of each drug per kilogram body mass into each mouse instead of the same mass into each mouse.

Explain why.

[3 marks]

0 5 . 4

Explain why the scientists include group 3 in the investigation.

[2 marks]



Figure 6 shows the results.

Figure 1: Survival of mice after treatment with Vincristine, DMSO or Resveratrol

The graph displays the percentage survival of mice over a 30-day period following treatment with Vincristine, DMSO, or Resveratrol. The y-axis represents 'Percentage survival' from 0 to 100, and the x-axis represents 'Time after treatment / days' from 0 to 30. The legend indicates that Vincristine is represented by squares, DMSO by triangles, and Resveratrol by circles.

Approximate data points extracted from the graph:

Time (days)	Vincristine (%)	DMSO (%)	Resveratrol (%)
0	100	100	100
8	100	100	95
10	100	80	95
12	100	80	80
14	100	65	75
16	95	55	70
18	80	25	30
20	70	5	15
22	50	5	5
24	35	0	0
26	20	0	0
28	10	0	0
30	0	0	0

Evaluate the use of vincristine and resveratrol in treating cancer.

Use information from the scientists' method and from the results in **Figure 6**.

[4 marks]

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[6 marks]

[illegible]

[6 marks]

[illegible]

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Although glucose enters and leaves the blood at variable rates, the concentration of glucose in the blood is kept between 4.4 and 6.6 mmol dm⁻³

[6 marks]

[illegible]

20

END OF QUESTIONS



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